

# [ Endpoint Monitoring, Process Monitoring, & FIM Essentials ]

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## [ 1. Endpoint Monitoring ]

### [ What is Endpoint Monitoring? ]

Endpoint Monitoring is the continuous observation of activities occurring on endpoints such as laptops, desktops, servers, and workstations to detect suspicious behavior and security threats.

### [ Telemetry Parameters: What is Monitored? ]

```
[ Endpoint Telemetry Data Streams ]
├─ System Execution  —> Process creation and termination
├─ Identity & Access  —> User logins and logouts
├─ File System Flow   —> File creation, modification, and deletion
├─ Network Activity   —> Active outbound/inbound network connections
├─ Physical Layer     —> USB device connectivity and activity
└─ OS Configuration  —> Registry changes (Windows) & Script executions (PowerShell/Cmd)
```

### [ Why is it Important? ]

It enables SOC teams to detect stealthy malware and ransomware, identify unauthorized system access, monitor anomalous user behaviors, and gather forensic timelines to support incident investigation.

### [ Incident Tracking Lifecycle ]

#### **Real-World Scenario: Malicious File Download**

When a threat actor breaches a system, Endpoint Monitoring continuously logs:

1. File download string extraction.
2. Process execution and thread spawning.
3. Outbound network socket creation back to the attacker server.
4. Persistence registry creation for reboot survival.

### [ SOC L1 Core Use Cases ]

- Investigating automated SIEM malware alerts.
  - Reviewing granular raw endpoint host timelines during triage.
  - Detecting unusual user behavior metrics (e.g., non-business hours commands).
  - Tracking attacker post-compromise activities.
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## [ 2. Process Monitoring ]

### [ What is Process Monitoring? ]

Process Monitoring is the systematic tracking and cryptographic behavioral analysis of running processes on an active system. A process is simply a running program in memory (Examples: `explorer.exe`, `chrome.exe`, `powershell.exe`, `cmd.exe`, `notepad.exe`).

## [ Forensic Variables Monitored ]

- Process Name & Binary File Path
- Process ID (PID) & Parent Process ID (PPID)
- Complete command-line arguments and switches
- Associated User Account and privilege context
- Exact millisecond start and stop timestamps
- Child processes spawned by the main binary thread

## [ Why is it Important? ]

Attackers frequently use legitimate, trusted administrative OS tools to perform malicious actions. This evasion methodology is known as Living-off-the-Land (LotL). Process monitoring is vital to detect the abuse of utilities like powershell.exe, cmd.exe, wscript.exe, rundll32.exe, and regsvr32.exe.

## [ Malicious Process Lineage Chain ]

```
[ Phishing Evasion Chain ]
  outlook.exe (Email Client)
    |
    ▼
  winword.exe (Office Application launching attachment)
    |
    ▼
  powershell.exe (Abused script shell executing encoded string)
    |
    ▼
  malware.exe (Hidden malicious binary execution)
```

## [ Suspicious Process Indicators ]

- Encoded or obfuscated base64 PowerShell script execution parameters.
- System shells (cmd.exe) spawning unusual binaries out of tmp storage directories.
- Standard Office applications directly spawning terminal shells or administrative tool windows.
- Multiple failed process executions occurring within a tight time window.

## [ SOC Investigation Checklist ]

1. What specific process binary started executing?
2. Which exact user account initiated the execution thread?
3. What was the originating parent process (PPID)?
4. What specific command line arguments and switches were passed?
5. Did the target process spawn further downstream child processes?

# [ 3. File Integrity Monitoring (FIM) ]

## [ What is File Integrity Monitoring? ]

File Integrity Monitoring (FIM) explicitly tracks changes made to critical files, system binaries, and production directories. It acts as an early indicator for unauthorized configuration modifications, automated malware activities, insider threats, and compliance posture violations.

## [ Scope of FIM Monitoring ]

```
[ FIM Event Classifications ]
├─ File Creation &————> New file added to monitored directory
├─ File Deletion &————> Critical security asset or log file removed
```

└─ File Modification —> Data or configuration inside file altered  
└─ Metadata Alterations -> File permission settings or ownership attributes changed

[ How Does It Work? ]

FIM calculations establish a baseline reference map (known good state) by hashing critical files. It then continuously re-hashes the assets to detect changes.

```
[ Hash Verification Loop ]
  Original config.txt  —> Hash Algorithm (SHA-256) —> Baseline: A1B2C3D4
                                                              | (Mismatch Alert!)
  Modified config.txt  —> Hash Algorithm (SHA-256) —> Current:  X9Y8Z7W6
```

Common cryptographic hash algorithms used include MD5, SHA-1, and SHA-256.

[ Real-World SOC Scenario ]

A threat actor gains access to a Linux system and attempts to alter local system identities by adding an unauthorized user into the primary password directory path (/etc/passwd). FIM immediately logs that the file hash, timestamp, and metadata fields changed, and fires a critical severity SIEM alarm to the triage queue.

[ SOC L1 FIM Investigation Steps ]

- 1. Identify the exact file path that triggered the change alert.
- 2. Identify the specific user account or process context that executed the modification.
- 3. Verify change management logs to ensure whether the alteration was authorized.
- 4. Review all concurrent processes active on that host during the modification time window.
- 5. Escalate immediately to the Incident Response team if unauthorized behavior is confirmed.

[ Quick Architecture Comparison ]

| Feature Metric              | Endpoint Monitoring                                    | Process Monitoring                                    | File Integrity Monitoring (FIM)                            |
|-----------------------------|--------------------------------------------------------|-------------------------------------------------------|------------------------------------------------------------|
| Core Analysis Focus         | Entire holistic endpoint activity framework.           | Running memory programs and binary execution paths.   | Storage configuration and critical file changes.           |
| Primary Detection Target    | Malware deployment, unauthorized interactive access.   | Malicious system process abuse, shell exploits.       | Unauthorized file modifications, ransomware encryption.    |
| Telemetry Data Source       | Unified OS security and connection logs.               | Kernel process lifecycle scheduling events.           | Filesystem kernel state and cryptographic hash audits.     |
| SOC Workflow Usage          | Incident triage, footprint scoping, timeline auditing. | Active threat detection, behavioral hunting patterns. | System hardening compliance, change validation monitoring. |
| Practical Indicator Example | Malicious file execution tracking.                     | PowerShell encoded argument flags abuse.              | Modified system infrastructure configuration files.        |